

ATHARV NASARE

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OBJECTIVE

Embedded / Firmware Engineer with hands-on experience in Embedded C, Linux, ESP32, STM32, and CAN communication. Experienced in RTOS-based firmware development, sensor interfacing, and debugging using embedded tools. Interested in working on embedded Linux systems, communication protocols, and automotive-grade software development.

SKILLS

Technical Skills	C, Embedded C, Shell Scripting, Python
Tools	GCC, GDB (Basic), Git, Keil uVision, STM32CubeIDE, KiCad, EasyEDA
Communication Protocols	CAN, UART, SPI, I2C, RS232
Embedded Platforms	STM32, ESP32, ESP32-S3, RISC-V Microcontrollers
Operating Systems	Linux (Basics), FreeRTOS (ESP-IDF)
Embedded Concepts	Bare Metal Programming, Sensor Interfacing, ADC, GPIO, Timers, Interrupts, RTOS Tasks
PCB Design	Schematic Capture, PCB Layout Design, Component Placement, Routing (KiCad, EasyEDA)
Hardware Debugging	Oscilloscope, Logic Analyzer (Basic), Multimeter
Networking	TCP/IP (Basic), IoT Cloud Integration (Firebase, Blynk)

Embedded Knowledge :

- Microcontroller Peripheral Programming (GPIO, ADC, Timers)
- CAN bus communication and message framing
- Interrupt handling and task scheduling in RTOS
- Sensor interfacing and analog signal acquisition
- Embedded firmware debugging and testing

PROJECTS

Smart Energy Management System using CAN Protocol

C-DAC Project

Tech: ESP32, Embedded C, ESP-IDF, CAN, SN65HVD230 CAN Transceiver, Firebase

- Designed and developed a distributed energy monitoring system using dual ESP32 microcontrollers.
- Implemented CAN-to-CAN communication using SN65HVD230 at 500 kbps with proper 120-ohm termination for reliable industrial communication.
- Developed sensor interfacing and ADC-based data acquisition using ACS712, ZMPT101B, and DS18B20 sensors.
- Implemented threshold-based fault detection and buzzer alert system.
- Designed and transmitted CAN frames with message filtering and validation.
- Integrated Wi-Fi and Firebase cloud for real-time remote monitoring and web dashboard visualization.

GitHub Repository: [Smart-Energy-Management-System](#)

Crop Monitoring System using IoT

B.Tech Graduation Project

Tech: Embedded C, Microcontroller, DHT11, Soil Moisture Sensor, Relay Module, Blynk IoT

- Designed and developed an IoT-based smart irrigation system for real-time monitoring of soil moisture, temperature, and humidity.
- Built a 240V AC to 5V DC regulated power supply using transformer and rectifier circuit.
- Interfaced DHT11 and soil moisture sensors with microcontroller for environmental monitoring.
- Implemented relay-based automatic motor control using threshold logic.
- Integrated Blynk platform for remote monitoring and motor control via mobile dashboard.

GitHub Repository: [Crop-Monitoring-System](#)

EDUCATION

(C-DAC) Sunbeam Institute Of Information Technology, Pune PG-Diploma in Embedded System Design	Aug 2025 – Feb 2026 <i>68.75%</i>
Bapurao Deshmukh College of Engineering, Sewagram, Wardha B.Tech in Electronics and Telecommunication Engineering, RTMNU	2021 – 2025 <i>CGPA: 7.2/10</i>
Janta Junior College, Deoli, Wardha HSC (12th), State Board	2019 – 2021 <i>82.5%</i>
School of Scholars, Sawangi (Meghe), Wardha SSC (10th), CBSE Board	2013 – 2019 <i>72.8%</i>

EXTRA-CURRICULAR ACTIVITIES

- Participation in technical and sports competitions , Dancing , Swimming

LANGUAGES

- English
- Hindi
- Marathi